

1.2 Classwork: Speed, Velocity, and Unit Conversion

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

1 mi = 5280 ft 1 mi $\approx$ 1.61 km 1 m $\approx$ 3.28 ft 1 ft = 12 in
1 km = 1000 m 1 in = 2.54 cm 1 h = 3600 s 1 min = 60 s

1. A student walks 24.0 meters down the hallway in 18.5 seconds.

(a) Find the average speed in meters per second, rounded to three significant figures.

$$S = \frac{24.0 \text{ m}}{18.5 \text{ s}} = 1.297297... \approx 1.30 \text{ m/s}$$

(b) How many significant figures does your answer have, and why not more?

3, The input values both have 3 sig figs.

2. Convert each speed. Write each conversion as a fraction and show the units cancelling.

(a) A bus on Via del Corso travels at 30.0 kilometers per hour. What is this speed in meters per second?

$$30 \frac{\text{km}}{\text{hr}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ hr}}{3600 \text{ s}} = 8.333... \approx 8.33 \text{ m/s}$$

(b) A train travels at 25.0 meters per second. What is this speed in kilometers per hour?

$$25.0 \frac{\text{m}}{\text{s}} \cdot \frac{1 \text{ km}}{1000 \text{ m}} \cdot \frac{3600 \text{ s}}{1 \text{ hr}} = 90.0 \frac{\text{km}}{\text{h}}$$

3. A fastball leaves the pitcher's hand at 95 miles per hour.

(a) What is this speed in feet per second?

$$95 \frac{\text{miles}}{\text{hr}} \cdot \frac{5280 \text{ ft}}{1 \text{ mile}} \cdot \frac{1 \text{ hr}}{3600 \text{ s}} = 139.33... \approx 139 \text{ ft/s}$$

(b) What is this speed in meters per second?

$$139.333... \frac{\text{ft}}{\text{sec}} \cdot \frac{1 \text{ m}}{3.28 \text{ ft}} = 42.4796... \approx 42.5 \text{ m/s}$$

* rounded
value

4. Three vehicles: a car at 60 miles per hour, a train at 95 kilometers per hour, and a motorcycle at 27 meters per second. Convert all three to meters per second, then list them from slowest to fastest.

$$\begin{aligned} \text{Car} & 60 \frac{\text{mi}}{\text{hr}} \cdot \frac{5280 \text{ ft}}{1 \text{ mile}} \cdot \frac{1 \text{ m}}{3.28 \text{ ft}} \cdot \frac{1 \text{ hr}}{3600 \text{ s}} = 26.8224 \dots \approx 26.8 \frac{\text{m}}{\text{s}} \\ \text{Train} & 95 \frac{\text{km}}{\text{hr}} \cdot \frac{1000 \text{ m}}{1 \text{ km}} \cdot \frac{1 \text{ hr}}{3600 \text{ s}} = 26.3888 \dots \approx 26.4 \frac{\text{m}}{\text{s}} \\ \text{Motorcycle} & 27 \frac{\text{m}}{\text{s}} \end{aligned}$$

train, car, motorcycle

5. The speed of sound in air is 343 meters per second. You see a lightning flash and hear the thunder 4.0 seconds later. How far away was the lightning, in meters? In miles?

$$343 \frac{\text{m}}{\text{s}} \cdot 4.0 \text{ s} = 1372 \text{ m} \approx 1400 \text{ m}$$

1.4 km

$$1.372 \text{ km} \cdot \frac{1 \text{ mile}}{1.61 \text{ km}} = 0.85217 \dots \approx 0.85 \text{ miles}$$

6. Our building is a New York brownstone, and the classroom is on the 6th floor — so from the entrance you climb 5 floors. Each floor is one flight of 18 steps, and each step is 8.0 inches high.

- (a) How many steps do you climb in total?

$$5 \text{ floor} \times \frac{18 \text{ steps}}{1 \text{ flr}} = 90 \text{ steps}$$

- (b) What is the total elevation gain, in feet? In meters?

$$90 \text{ steps} \times \frac{8.0 \text{ inches}}{1 \text{ step}} \times \frac{1 \text{ ft}}{12 \text{ in}} = 60 \text{ ft}$$

$$\begin{aligned} 60 \text{ ft} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \\ = 18.29 \\ \approx 18 \text{ m} \end{aligned}$$

- (c) Climbing steadily at 1.2 steps per second, how long does the climb take? Give your answer in seconds, and again in minutes and seconds.

$$90 \text{ steps} \cdot \frac{1 \text{ sec}}{1.2 \text{ steps}} = 75 \text{ s} \quad \text{or } 1 \text{ min}, 15 \text{ sec}$$

- (d) What is your average vertical speed, in feet per minute? In meters per second?

$$S = \frac{60 \text{ ft}}{1.25 \text{ min}} = 48 \text{ ft/min} \quad \text{or} \quad \frac{18.29 \text{ m}}{75 \text{ s}} = 0.24386 \dots \approx 0.24 \frac{\text{m}}{\text{s}}$$

7. Challenge: a car's speedometer reads 30 miles per hour. A student converts this to kilometers per hour and writes 48.28032 km/h. Explain what is wrong with this answer, and write a better one.

It is too precise. It should be at most two significant figures: 48 km/h